



Relations and Their Properties

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What is a Relation?

A **binary relation R** from set **A** to set **B** is a subset of the Cartesian product $A \times B$. It specifies which elements of A are 'related to' which elements of B.

Definition and Notation

$$R \subseteq A \times B$$

$(a, b) \in R$ means a is related to b

Relation ON a set A:

$R \subseteq A \times A$ (both elements from same set)

Example (on $A = \{1, 2, 3\}$):

$R = \text{"less than"} = \{(1,2), (1,3), (2,3)\}$

Representations:

- Set of ordered pairs
- Boolean matrix (1 if related, 0 if not)
- Arrow (directed graph) diagram

Concrete Example

Let $A = \{1, 2, 3\}$ and $R = \text{"divides"}$

aRb means a divides b evenly

Pairs in R:

$(1,1)$: 1 divides 1 ✓

$(1,2)$: 1 divides 2 ✓

$(1,3)$: 1 divides 3 ✓

$(2,2)$: 2 divides 2 ✓

$(3,3)$: 3 divides 3 ✓

$(2,3)$: 2 divides 3 ✗ (not in R)

$R = \{(1,1), (1,2), (1,3), (2,2), (3,3)\}$

Five Key Properties of Relations

These properties are defined for a relation R on a set A :

Property	Formal Definition	Real Example	Counter-example
Reflexive	$\forall a \in A: (a,a) \in R$	" $\leq$ " on integers ($1 \leq 1, 2 \leq 2, \dots$)	" $<$ " on integers ($1 < 1$ is false)
Irreflexive	$\forall a \in A: (a,a) \notin R$	" $<$ " on integers (no $a < a$)	" $\leq$ " includes (a,a) pairs
Symmetric	$(a,b) \in R \Rightarrow (b,a) \in R$	"sibling of" among people	"parent of" — not symmetric
Antisymmetric	$(a,b) \in R \wedge (b,a) \in R \Rightarrow a=b$	" $\leq$ " on integers	"sibling of" (both aRb and bRa , $a \neq b$)
Transitive	$(a,b) \in R \wedge (b,c) \in R \Rightarrow (a,c) \in R$	" $<$ " on integers (if $a < b$ and $b < c$ then $a < c$)	"parent of" — not transitive

Equivalence Relations

A relation R on set A is an **Equivalence Relation** if and only if it is **Reflexive, Symmetric, AND Transitive** — all three together.

1. Reflexive

$$\forall a \in A: (a,a) \in R$$

2. Symmetric

$$(a,b) \in R \Rightarrow (b,a) \in R$$

3. Transitive

$$(a,b), (b,c) \in R \Rightarrow (a,c) \in R$$

Classic Example: Congruence Modulo n

Define: $a \equiv b \pmod{n}$ iff n divides $(a - b)$

Example with $n = 3$:

$$7 \equiv 1 \pmod{3} \text{ because } 3 \text{ divides } (7-1) = 6 \quad \checkmark$$

$$10 \equiv 4 \pmod{3} \text{ because } 3 \text{ divides } (10-4) = 6 \quad \checkmark$$

Equivalence Classes:

$$[0] = \{\dots, -6, -3, 0, 3, 6, 9, \dots\} \quad [1] = \{\dots, -5, -2, 1, 4, 7, 10, \dots\} \quad [2] = \{\dots, -4, -1, 2, 5, 8, 11, \dots\}$$

Key Theorem: An equivalence relation partitions the set into disjoint equivalence classes.

Partial Orderings

A **Partial Ordering** is a relation R on set A that is **Reflexive, Antisymmetric, AND Transitive**. The pair (A, R) is called a partially ordered set (POSET).

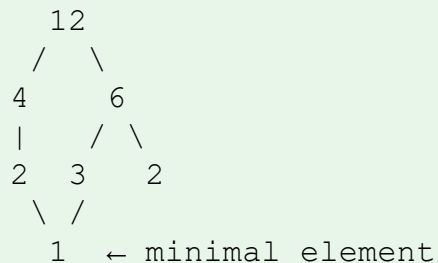
Example: Divisibility on $\{1, 2, 3, 4, 6, 12\}$

$a \mid b$ means a divides b

Verify:

- ✓ Reflexive: $a \mid a$ (every number divides itself)
- ✓ Antisymmetric: if $a \mid b$ and $b \mid a$, then $a = b$
- ✓ Transitive: if $a \mid b$ and $b \mid c$, then $a \mid c$

Hasse Diagram (removes reflexive & transitive edges):



Key Terminology

Minimal Element:

No element precedes it.

In divisibility: 1 (nothing divides 1 except 1 itself)

Maximal Element:

No element comes after it.

In divisibility on $\{1..12\}$: 12

Total Order (Linear Order):

Every pair is comparable.

Example: $\leq$ on integers

Partial Order:

Some pairs may be incomparable.

e.g., 4 and 6 in divisibility: neither $4 \mid 6$ nor $6 \mid 4$.

Applications in Computer Science

Relations are fundamental data structures underlying many CS systems:

Relational Databases

Every table in a relational database IS a relation (set of tuples). SQL joins combine relations using Cartesian product and selection. Database normalization is based on functional dependencies — a special type of relation.

Social Networks

'Friend', 'follows', and 'blocks' are all relations on a set of users. Symmetric relations (friend) vs. asymmetric (follow). Transitivity of 'knows' gives 'Six degrees of separation'. Equivalence classes group users in communities.

Compiler Design

Variable dependencies in code form a relation. Computing the transitive closure finds all indirect dependencies. Type compatibility (subtype relation) is a partial order. Operator precedence is a partial ordering of operators.

Task Scheduling

Project tasks have 'must complete before' constraints — a partial order (DAG). Topological sort of the Hasse diagram gives a valid execution order. Critical path analysis uses this POSET structure to minimize project duration.



THANK YOU

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